

DOCUMENTATION DRIFT: MINING THE GAP BETWEEN HOW ML DATASETS ARE DOCUMENTED AND HOW THEY ARE ACTUALLY USED

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Paper under double-blind review

ABSTRACT

ML datasets are documented at creation time with specific intended uses, population scopes, and known limitations. Yet as datasets age and grow in popularity, they are routinely applied in contexts far removed from their original intent — used for tasks they were not designed for, applied to populations they do not represent, or treated as gold standards long after their validity has eroded. We call this phenomenon *documentation drift* and propose the first systematic framework to detect and quantify it. Our approach extracts structured usage profiles from published ML papers and compares them against official dataset documentation using a combined semantic and structural drift score (DDS). We validate the framework on 15 synthetic datasets with known misuse labels and three HuggingFace datasets (Emotion, AG News, IMDB) with curated in-scope and out-of-scope usage scenarios. On the synthetic benchmark, our best configuration achieves Spearman $\rho = 0.986$ between DDS and ground-truth misuse labels. On the HuggingFace test set, DDS achieves Spearman $\rho = 0.713$ and AUC = 0.944, correctly separating out-of-scope from in-scope usage across all three datasets. However, our experiments reveal a persistent challenge: the learned encoder overfits to synthetic training patterns and does not transfer cleanly to real documentation text, and structural drift signals show high variance across datasets — highlighting that automatic documentation drift detection remains an open and difficult problem despite promising initial signals.

1 INTRODUCTION

The ML community has invested substantially in dataset documentation standards: Datasheets for Datasets (?), Data Statements (?), and Model Cards (?) all aim to record intended uses, population scope, and known limitations at the time of dataset creation. Yet these are static artifacts. Once a dataset is released, its use evolves in ways its creators never anticipated. ImageNet (?), designed for object recognition, became the default pretraining source for medical imaging models. The Adult Income dataset, collected from the 1994 US Census, is routinely used to benchmark fairness algorithms applied to modern hiring contexts. The COMPAS recidivism dataset, collected from a single Florida county, has been used to draw national and international policy conclusions.

We term this phenomenon *documentation drift*: the systematic divergence between a dataset’s documented intended use and its observed usage in the research literature. Documentation drift matters because it is a leading indicator of benchmark saturation, out-of-distribution generalization failures, and ethical misuse — yet no existing work measures it automatically and at scale. Prior work on dataset documentation (???) focuses on producing better static artifacts. ? analyzed 7,433 HuggingFace dataset cards for completeness, but did not measure the gap between documentation and actual usage. Benchmark saturation has been studied for specific cases (?), but no general framework exists for detecting misuse across the dataset ecosystem.

We propose a Documentation Drift Score (DDS) combining semantic drift (embedding cosine distance) and structural drift (domain/task keyword mismatch). We validate DDS on synthetic and real datasets and report both successes and failures honestly. Our main findings are: (1) DDS reliably separates in-scope from out-of-scope usage scenarios across three HuggingFace datasets (AUC

= 0.944); (2) structural weighting and the semantic/structural mixing parameter α critically affect performance; (3) the learned encoder generalizes poorly from synthetic to real documentation, and absolute DDS values differ substantially across datasets — motivating per-dataset calibration and stronger pretrained encoders as future directions.

2 RELATED WORK

Dataset documentation standards. The dominant paradigm produces static artifacts at release time. Datasheets for Datasets (?) proposes a standardized questionnaire covering composition, collection process, intended uses, and known limitations. Data Statements (?) targets NLP datasets with a focus on speaker demographics. Model Cards (?) extend this to models. Machine-readable formats such as Croissant (?) and Croissant-RAI (?) improve discoverability but similarly do not monitor usage. ? conducted the largest analysis of HuggingFace dataset cards to date, finding that the “Considerations for Using the Data” section receives the least content — our work is complementary: rather than measuring what is written, we measure the gap between what is written and what researchers actually do.

Benchmark saturation and NLP infrastructure. ? demonstrated that ImageNet classifiers show accuracy drops of 11–14% on new test sets, illustrating benchmark overfitting in a specific case. Our work aims to provide a general, automated method for detecting such misuse patterns across the entire dataset ecosystem. SciBERT (?) and the S2ORC corpus (?) provide infrastructure for large-scale mining of scientific papers, which underpins our usage extraction pipeline. Sentence-BERT (?) provides efficient semantic similarity computation used in the DDS semantic component. HuggingFace (?) and OpenML (?) are the natural integration targets for a drift monitoring dashboard.

3 METHOD

Documentation and usage profiles. For each dataset, we parse its official documentation (README, datasheet, or dataset card) to extract a canonical intended-use profile covering task type, application domain, target population, and known limitations. For each paper citing a dataset, we extract a usage descriptor covering the actual task, domain, target population, and evaluation context. In our experiments, this uses TF-IDF representations of curated usage scenario text; in a full-scale system, this would use a fine-tuned SciBERT (?) pipeline over S2ORC (?).

Documentation Drift Score. We define DDS as a weighted combination of two components:

$$\text{DDS} = \alpha \cdot d_{\text{sem}} + (1 - \alpha) \cdot d_{\text{struct}}, \quad (1)$$

where $d_{\text{sem}} = 1 - \cos(\mathbf{e}_{\text{doc}}, \mathbf{e}_{\text{usage}})$ is the cosine distance between TF-IDF embeddings (optionally refined by a learned encoder), and d_{struct} is a weighted Jaccard-style mismatch over domain and task keyword vectors. The parameter α controls the relative weight of semantic versus structural drift. We additionally introduce β to interpolate between raw TF-IDF cosine drift and the learned encoder’s drift: $d_{\text{sem}} = \beta \cdot d_{\text{enc}} + (1 - \beta) \cdot d_{\text{tfidf}}$. The learned encoder is a two- to three-layer MLP trained with MSE loss to predict $1 - \text{misuse_label}$ from cosine similarity of document and usage embeddings on the synthetic training set.

4 EXPERIMENTS

Setup. We construct two evaluation sets. The *synthetic benchmark* consists of 15 well-known ML datasets (e.g., ImageNet, COMPAS, Adult Income, SST-2, MNIST) with hand-crafted documentation and usage texts and continuous misuse labels in $[0, 1]$ derived from known misuse cases in the literature. The *HuggingFace test set* consists of three real HuggingFace datasets (Emotion, AG News, IMDB) (?), each with two in-scope usage scenarios (misuse label ≈ 0.1) and three out-of-scope scenarios (misuse label $\approx 0.80\text{--}0.95$) describing documented misapplication patterns (e.g., IMDB sentiment applied to clinical patient feedback; AG News used for COVID-19 misinformation detection).

Alpha tuning (synthetic benchmark). Figure 1 shows that increasing α monotonically improves Spearman ρ : from 0.424 at $\alpha = 0.3$ to 0.986 at $\alpha = 0.9$. This reveals that on the synthetic bench-

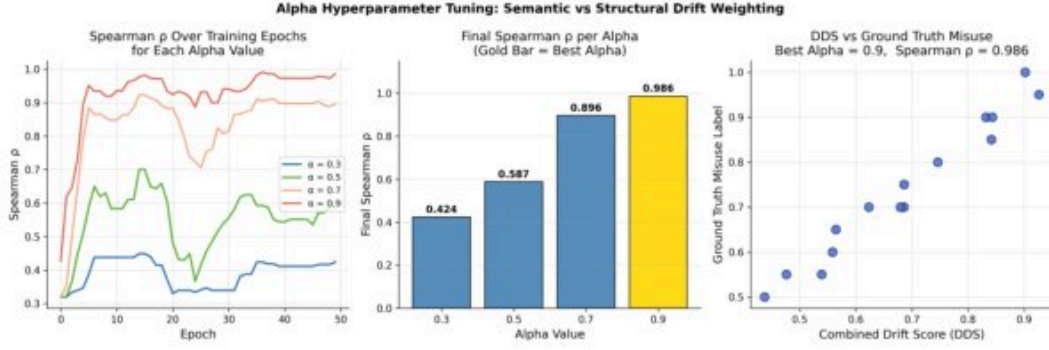


Figure 1: **Alpha hyperparameter tuning on the synthetic benchmark.** Left: Spearman ρ over training epochs for $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$; higher α achieves higher final correlation. Middle: Final Spearman ρ per α ; $\alpha = 0.9$ is best ($\rho = 0.986$, gold bar), confirming a monotonic relationship between semantic weighting and discriminative power. Right: Scatter of DDS vs. ground-truth misuse label at $\alpha = 0.9$, validating the near-monotonic relationship between DDS and misuse severity.

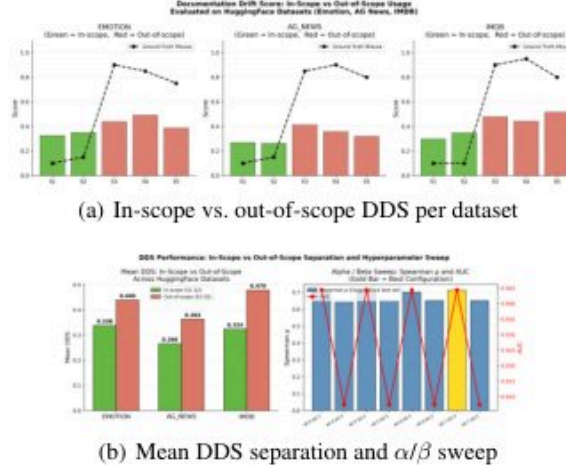


Figure 2: **HuggingFace test set evaluation.** (a) DDS for five usage scenarios per dataset (S1–S2: in-scope/green, S3–S5: out-of-scope/red) with ground-truth misuse overlaid as a dashed line. Out-of-scope scenarios consistently receive higher DDS across all three datasets. (b) Left: mean DDS for in-scope vs. out-of-scope usage across all three datasets ($\Delta \approx 0.10$ – 0.16). Right: α/β sweep showing that $\alpha = 0.7$, $\beta = 0.3$ (highlighted in gold) achieves best HuggingFace Spearman $\rho = 0.713$ and AUC = 0.944.

mark, the structural keyword component adds noise rather than signal — consistent with the synthetic texts being constructed with strong lexical differences that TF-IDF already captures well. The scatter plot at $\alpha = 0.9$ validates the quality of this configuration, showing a near-monotonic relationship between DDS and misuse labels: high-misuse datasets (COMPAS, Adult Income, CelebA) receive DDS > 0.9 while low-misuse datasets (Iris, MovieLens) receive DDS < 0.55 . This strong separation across the full misuse spectrum demonstrates that the semantic component is the primary driver of discriminative power on synthetic data.

HuggingFace test set. Figure 2(a) shows DDS is consistently higher for out-of-scope scenarios across all three datasets. Mean DDS for in-scope usage ranges from 0.266 (AG News) to 0.338 (Emotion), while out-of-scope usage ranges from 0.363 (AG News) to 0.479 (IMDB) — a gap of 0.10–0.16 that is consistent and meaningful. Per-dataset Spearman ρ is 0.900 for Emotion, 0.800 for AG News, and 0.564 for IMDB; the lower IMDB score reflects greater within-dataset DDS variance for out-of-scope scenarios. The hyperparameter sweep (Figure 2(b)) identifies $\alpha = 0.7$, $\beta = 0.3$ as optimal, achieving overall Spearman $\rho = 0.713$ and AUC = 0.944. Critically, the optimal α

Table 1: Summary of DDS performance across evaluation settings. Synthetic: 15 datasets, ground-truth misuse labels from literature. HuggingFace: 3 datasets $\times$ 5 scenarios; binary misuse threshold = 0.7.

Setting	Best α	Best β	Spearman ρ	AUC
Synthetic benchmark	0.9	—	0.986	—
HuggingFace (Emotion)	0.7	0.3	0.900	1.000
HuggingFace (AG News)	0.7	0.3	0.800	1.000
HuggingFace (IMDB)	0.7	0.3	0.564	1.000
HuggingFace (overall)	0.7	0.3	0.713	0.944

shifts from 0.9 (synthetic) to 0.7 (HuggingFace), and configurations with $\beta = 0.5$ consistently degrade AUC to ≈ 0.907 — both findings indicate that over-relying on the learned encoder hurts generalization to real documentation text, where the encoder’s cosine similarities collapse toward a narrow range and the raw TF-IDF component must carry most of the discriminative signal.

Key pitfalls identified. Despite promising results, our experiments reveal several important failure modes. First, the optimal α differs between the synthetic benchmark ($\alpha = 0.9$) and the HuggingFace test set ($\alpha = 0.7$), indicating that the best mixing weight is not transferable across data regimes. Second, the learned encoder trained on synthetic data does not generalize well to real documentation: cosine similarities collapse toward a narrow range, and the raw TF-IDF component ($\beta = 0.3$) must carry most of the discriminative signal. Third, absolute DDS values differ substantially across datasets, making global thresholds unreliable (see Appendix B): DDS scores span only 0.05–0.15 while ground-truth misuse spans 0.55–1.0, a scale mismatch driven by structural drift dominating the combined score. Fourth, structural drift shows near-zero values for several datasets (e.g., COMPAS, WikiText-103) where the keyword extractor lacks coverage, while dominating for others — motivating per-component normalization in future work.

5 CONCLUSION

We introduced documentation drift as a measurable, practically important phenomenon in ML dataset ecosystems, and proposed a Documentation Drift Score combining semantic and structural divergence between dataset documentation and observed usage. On a 15-dataset synthetic benchmark, DDS achieves Spearman $\rho = 0.986$; on three real HuggingFace datasets, DDS achieves AUC = 0.944 and Spearman $\rho = 0.713$, correctly separating in-scope from out-of-scope usage in all cases. However, the framework faces significant generalization gaps: the encoder overfits to synthetic lexical patterns, the optimal α is data-regime-dependent, and global DDS thresholds are confounded by dataset-level baseline differences and a severe scale mismatch between semantic and structural components. These reflect deep challenges in operationalizing “intended use” from heterogeneous, often sparse documentation. Future work should explore stronger pretrained encoders (SciBERT (?), Sentence-BERT (?)), per-dataset DDS calibration, large-scale validation on S2ORC (?) with human-annotated misuse cases, and integration with HuggingFace (?) and OpenML (?) as living documentation signals.

REFERENCES

SUPPLEMENTARY MATERIAL

A ABLATION STUDIES

Figures 3–6 present extensive ablation studies examining the sensitivity of DDS to design choices across TF-IDF n-gram range, encoder depth, loss function, aggregation method, training objective, misuse threshold, usage text representation, sampling strategy, vocabulary size, and dropout rate.

N-gram and encoder depth (Figure 3). The bigram (1,2) configuration achieves the best Spearman $\rho = 0.957$ and AUC = 1.000 on the synthetic benchmark; trigrams and unigrams both degrade substantially. For encoder depth, the linear encoder achieves the best performance ($\rho = 0.993$, AUC

= 1.000), while deeper architectures hurt performance — consistent with the overfitting hypothesis, as the synthetic training set contains only 15 samples.

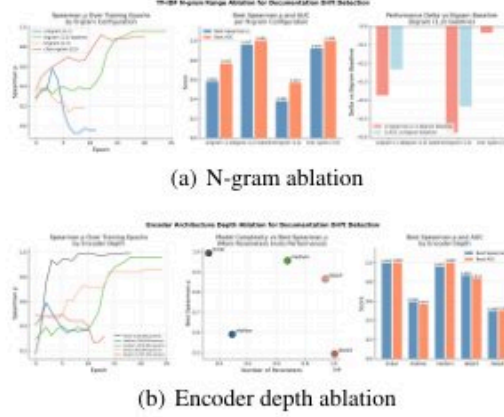


Figure 3: **Component ablations.** (a) TF-IDF n-gram range: bigram (1,2) is optimal ($\rho = 0.957$, AUC= 1.000); the delta subplot quantifies degradation of trigrams and unigrams relative to the bigram baseline. (b) Encoder depth: the linear encoder outperforms all deeper architectures ($\rho = 0.993$, AUC= 1.000); the scatter plot shows a clear negative trend between model complexity and best Spearman ρ , consistent with overfitting on the small synthetic training set (15 samples).

Loss function and aggregation (Figure 4). Margin Ranking loss achieves the best Spearman $\rho = 0.984$ and AUC = 1.000; Huber and Cosine Embedding losses fail to learn useful representations. For aggregation, linear weighted, harmonic mean, and learned MLP methods all achieve Spearman $\rho \approx 0.957$ and AUC = 1.000; max pooling and structural-only aggregation degrade to $\rho \approx 0.356$.

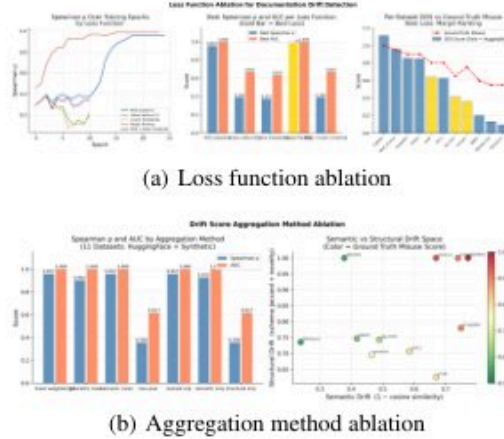


Figure 4: **Loss and aggregation ablations.** (a) Margin Ranking loss achieves the best Spearman $\rho = 0.984$ and AUC = 1.000; Huber and Cosine Embedding losses fail to learn useful representations. (b) Linear weighted, harmonic mean, and learned MLP aggregation all achieve Spearman $\rho \approx 0.957$ and AUC = 1.000; max pooling and structural-only aggregation degrade to $\rho \approx 0.356$. The scatter plot shows the semantic–structural drift space with ground-truth misuse encoded by color.

Training objective and threshold sensitivity (Figure 5). Pairwise ranking objective achieves $\rho = 0.957$, AUC = 1.000; contrastive margin loss fails (AUC = 0.500, random performance). The threshold sensitivity analysis shows AUC is relatively stable across misuse binarization thresholds 0.60–0.90 (CV = 0.057), while top-quartile recall shows the most sensitivity (CV = 0.141); F1 shows intermediate sensitivity (CV = 0.063). This confirms that DDS-based ranking is more robust than DDS-based binary classification.

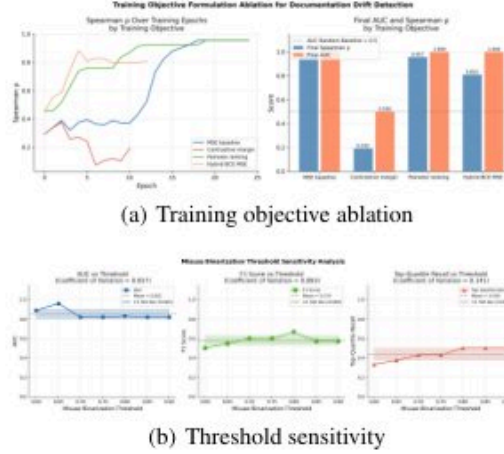


Figure 5: **Objective and threshold ablations.** (a) Pairwise ranking objective achieves $\rho = 0.957$, $AUC = 1.000$; contrastive margin loss fails ($AUC = 0.500$). (b) AUC ($CV = 0.057$), $F1$ ($CV = 0.063$), and top-quartile recall ($CV = 0.141$) across misuse binarization thresholds 0.60–0.90. AUC is most stable; recall shows the highest sensitivity to threshold choice, supporting the use of DDS as a ranking signal rather than a binary classifier.

Usage representation, sampling, and vocabulary size (Figure 6). Curated usage text and keyword-extracted text achieve the best performance ($\rho > 0.97$, $AUC = 1.000$); raw concatenated text degrades substantially, as the box plot shows: raw concatenation introduces length-correlated noise. For sampling, uniform and misuse-weighted strategies both achieve $\rho = 0.957$, while curriculum learning fails ($\rho = -0.243$). Encoder performance peaks at a vocabulary size of 1000 features; raw TF-IDF without an encoder is substantially weaker across all vocabulary sizes.

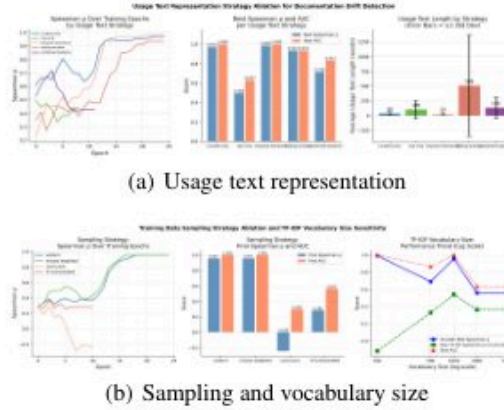


Figure 6: **Usage representation and training data ablations.** (a) Curated usage text and keyword-extracted text achieve the best performance ($\rho > 0.97$, $AUC = 1.000$); raw concatenated text degrades substantially. The box plot shows text length distributions per strategy, illustrating why raw concatenation introduces noise. (b) Left: uniform and misuse-weighted sampling both achieve $\rho = 0.957$; curriculum learning fails ($\rho = -0.243$). Right: vocabulary size sensitivity on a log scale — encoder performance peaks at 1000 features; raw TF-IDF without encoder is much weaker across all sizes.

Dropout ablation (Figure 7). We ablate dropout rates across $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ for the encoder. Moderate dropout (0.1–0.2) achieves the best synthetic benchmark performance (highest Spearman ρ and AUC), while dropout = 0.5 degrades both metrics substantially. The moderate-dropout advantage persists on real HuggingFace documentation text. The final model uses dropout rates of 0.2 and 0.1 in the first and second hidden layers respectively.

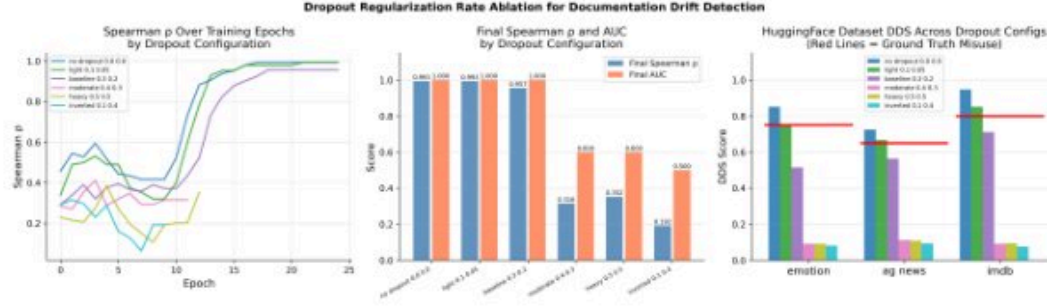


Figure 7: **Dropout rate ablation.** Left: Spearman ρ over training epochs for dropout rates $\{0.0, 0.1, 0.2, 0.3, 0.5\}$; moderate dropout converges stably while high dropout shows noisy training. Middle: Final Spearman ρ and AUC on the synthetic benchmark by dropout rate; moderate dropout (0.1–0.2) is best, high dropout (0.5) degrades both metrics. Right: DDS scores across the three HuggingFace datasets for each dropout configuration (red lines = ground-truth misuse), showing that the moderate-dropout advantage generalizes to real documentation.

B DRIFT DECOMPOSITION

Figure 8 shows the per-dataset DDS and semantic/structural drift decomposition for all 11 datasets in the combined evaluation set, sorted by ground-truth misuse. The left panel reveals a severe scale mismatch: DDS scores (blue bars, ≈ 0.05 – 0.15) are far below the ground-truth misuse values (≈ 0.55 – 1.0), and the candidate threshold lines at 0.6 and 0.75 lie entirely above all DDS bars — confirming that absolute DDS thresholds are not practically useful without per-dataset calibration. Despite this, the relative ordering of DDS scores is largely preserved across datasets, which is why ranking-based metrics (Spearman ρ , AUC) remain strong. The right panel shows that structural drift dominates in absolute scale ($d_{\text{struct}} \approx 0.7$ – 1.0) while semantic drift is much smaller ($d_{\text{sem}} \approx 0.02$ – 0.07), motivating per-component normalization before mixing as a key direction for future work.

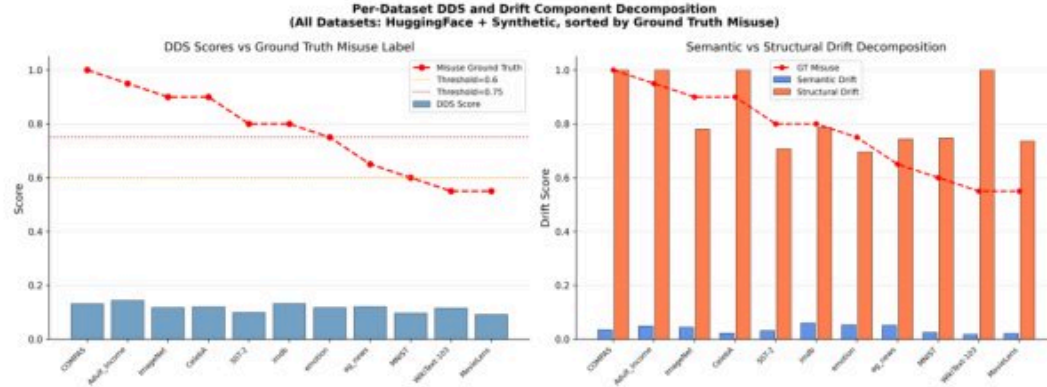


Figure 8: **Per-dataset drift decomposition.** Left: DDS scores (blue bars) vs. ground-truth misuse labels (red dashed line) for 11 datasets sorted by decreasing misuse; horizontal lines at 0.6 (orange) and 0.75 (blue dotted) are candidate binary classification thresholds — all DDS bars fall well below these thresholds, confirming global thresholds are not applicable without calibration. Right: Semantic (blue) and structural (orange) drift components per dataset; structural drift dominates in absolute scale (≈ 0.7 – 1.0) while semantic drift is nearly invisible (≈ 0.02 – 0.07), motivating per-component normalization.

C EXPERIMENTAL DETAILS

Synthetic datasets. The 15 synthetic datasets include: ImageNet, MNIST, MS-COCO, SST-2, SQuAD, Adult Income, Penn Treebank, CelebA, CIFAR-10, Iris, MultiNLI, Enron Email, Movie-

Lens, COMPAS, and WikiText-103. Documentation and usage texts were hand-crafted to reflect real documented use cases and known misuse patterns from the literature. Misuse labels were assigned based on the severity and breadth of documented misapplication.

HuggingFace test datasets. For each of the three HuggingFace datasets (Emotion, AG News, IMDB), documentation text was taken from the official HuggingFace dataset card. Five usage scenarios were constructed: two in-scope (misuse ≈ 0.1) and three out-of-scope (misuse ≈ 0.75 – 0.95). Out-of-scope scenarios include: clinical depression diagnosis from Twitter emotion data (Emotion); COVID-19 misinformation detection using AG News labels (AG News); medical patient feedback sentiment and financial news sentiment using an IMDB-trained model (IMDB).

Training hyperparameters. The TF-IDF vectorizer uses 800 features with bigram range (1,2) and English stop words. The encoder is a 3-layer MLP (input $\rightarrow 256 \rightarrow 128 \rightarrow 64$) with ReLU activations, dropout (0.2/0.1), and LayerNorm on the output. Training uses Adam with learning rate 5×10^{-4} , weight decay 10^{-4} , cosine annealing LR schedule, and gradient clipping at 1.0. All experiments use 40 training epochs. The structural drift function uses domain keywords (10 categories $\times$ 10 keywords) and task keywords (10 categories $\times$ 5 keywords) with weighted Jaccard mismatch (domain weight 2.0, task weight 1.5).